
Hands-On Projects



Project 2-1: Use an Online Password Cracker

In this project, you will create a digest on a password and then crack it with an online cracking website to demonstrate the speed of cracking passwords.

1. The first step is to use a hash algorithm to create a password digest. Use your web browser to go to www.fileformat.info/tool/hash.htm (if you

are no longer able to access the site through the web address, use a search engine to search for “Fileformat.Info hash functions”).

2. Under **String hash**, enter the simple password **apple123** in the **Text:** line.
3. Click **Hash**.
4. Scroll down the page and copy the MD4 hash of this password to your Clipboard by selecting the text, right-clicking, and choosing **Copy**.
5. Open a new tab on your web browser.
6. Go to <https://crackstation.net/>.
7. Paste the MD4 hash of *apple123* into the text box beneath **Enter up to 10 non-salted hashes:**.
8. In the RECAPTCHA box, enter the current value being displayed in the box that says **Type the text**.
9. Click **Crack Hashes**.
10. How long did it take this online rainbow table to crack this hash?
11. Click the browser tab to return to FileFormat.Info.
12. Under **String hash**, enter the longer password **12applesauce** in the **Text:** line.
13. Click **Hash**.
14. Scroll down the page and copy the MD4 hash of this password to your Clipboard.
15. Click to browser tab to return to the CrackStation site.
16. Paste the MD4 hash of *12applesauce* into the text box beneath **Enter up to 10 non-salted hashes:**.
17. In the RECAPTCHA box, enter the current value being displayed in the box that says **Type the text**.
18. Click **Crack Hashes**.
19. How long did it take this online rainbow table to crack this stronger password hash?
20. Click the browser tab to return to FileFormat.Info and experiment by entering new passwords, computing their hash, and testing them in the CrackStation site. If you are bold, enter a string hash that is similar to a real password that you use.
21. What does this tell you about the speed of cracking passwords? What does it tell you about how easy it is for attackers to crack weak passwords?
22. Close all windows.



Project 2-2: Download and Install a Password Management Application

The drawback to using strong passwords is that they can be very difficult to remember, particularly when a unique password is used for each account that a user has. As another option, password management applications allow users

to store account information such as a username and password. These programs are themselves protected by a single strong password. One example of a password storage application is KeePass Password Safe, which is an open-source product. In this project, you will download and install KeePass.



1. Use your web browser to go to **keepass.info** and then click **Downloads** (if you are no longer able to access the site through the web address, use a search engine to search for “KeePass”).
2. Under **Professional Edition**, locate the most recent portable version of KeePass and click it to download the application. Save this file in a location such as your desktop, a folder designated by your instructor, or your portable USB flash drive. When the file finishes downloading, install the program. Accept the installation defaults.



Because this is the portable version of KeePass it does not install under Windows. In order to use it, you must double-click the filename KeePass.exe.

3. Launch KeePass to display the opening screen.
4. Click **File** and **New** to start a password database. Enter a name for the password database and save it to your computer. Enter a strong master password for the database to protect all of the passwords in it. Enter the password again in the Repeat password box. Then, click **OK** twice.
5. Click **Edit** and **Add Entry**. You will enter information about an online account that has a password that you already use. Click **OK** when you are finished.
6. Create a group by clicking **Edit** and then **Add Group** and then enter **Web Sites**. Click **OK** when you are finished.
7. Select the **Web Sites** group and click **Edit** and then **Add Entry**.
8. Enter a title for your website (such as *Google Gmail*) under **Title**.
9. Under **User name**, enter the username that you use to log in to this account.
10. Erase the entries under **Password** and **Repeat** and enter the password that you use for this account and confirm it.
11. Enter the URL for this account under **URL**.
12. Click **OK**.
13. Click **File** and **Save**.
14. Exit KeePass.
15. If necessary, navigate to the location of KeePass and double-click the file **KeePass.exe** to launch the application.
16. Enter your master password to open your password file.

17. If necessary, click the group to locate the account you just entered; it will be displayed in the right pane.
18. Click the link next to the URL to go to that website.
19. Click KeePass in the taskbar so that the window is now on top of your browser window.
20. Drag and drop your username from KeePass into the login username box for this account in your web browser.
21. Drag and drop your password from KeePass for this account.
22. Click the button on your browser to log in to this account.
23. Because you can drag and drop your account information from KeePass, you do not have to memorize any account passwords and can instead create strong passwords for each account. Is this an application that would help users create and use strong passwords? What are the strengths of such password programs? What are the weaknesses? Would you use KeePass?
24. Close all windows.



Project 2-3: Download and Install an Online Vault Password Manager

One of the drawbacks to using a password management program like KeePass is that it must be launched whenever a password must be retrieved or the program must be left open, which could be a security risk. An option is to use a browser-based online vault password manager program that retrieves the passwords automatically. One example of a browser-based password storage program is LastPass, which enables you to access your passwords from any computer. In this project, you will download and install LastPass.

1. Use your web browser to go to **lastpass.com** and click **Download Free** (if you are no longer able to access the site through the web address, use a search engine to search for “LastPass”).
2. Click **Watch screencast tutorials to learn the basics**.
3. Click **Getting started with LastPass** to open the tutorial screen, and if necessary click the **Play** button in the middle of the screen.
4. When the Basic Instructions tutorial has completed, click your browser’s **Back** button.
5. Click **Watch screencast tutorials to learn the basics** again.
6. Click **How to Automatically Fill Webpage Forms** to open the tutorial screen, and if necessary click the **Play** button in the middle of the screen.
7. When the tutorial has completed, click your browser’s **Back** button.
8. Click the **Download** button to download LastPass.

9. After the program has downloaded, launch the program and follow the instructions for the default installation.
10. Under **Create or Log In** be sure to click **Create a New Account**.
11. Enter your email address and create a password. Be sure to remember this information.
12. Accept the default settings to finish installing and creating your LastPass account.



The steps may vary for installing LastPass depending upon which web browser you are using.

13. Close your web browser.



Project 2-4: Using a Browser-Based Password Management Program

In this project, you will use the LastPass program installed in the previous project.

1. Launch your web browser. If necessary, enable the LastPass browser extension.
2. Notice that you now have a LastPass button at the top of the screen. Click **LastPass**.
3. Enter your Master Password and then click **Login**.
4. Point your web browser to a website you frequently use that requires you to enter your username and password.
5. Enter your username and password. Notice that LastPass now asks if you want it to remember this password. Click **Save Site**.
6. When the **Add Site** window opens, click **Save**.
7. Log out of the website.
8. Now log in to two other websites and record their passwords in LastPass.
9. Close the web browser.
10. Reopen the web browser and click the **LastPass** icon on the toolbar. Notice that you are still logged in.
11. Click the **LastPass** icon and select the site that you want to visit. What happens when you go to these sites?



Your LastPass passwords can be retrieved from any other computer's web browser that has LastPass installed; you are not restricted to only this computer.

12. Because your login information automatically appears in LastPass, you do not have to memorize any account passwords and can instead create strong passwords for each account. Is this an application that would help users create and use strong passwords? What are the strengths of browser-based password program? What are the weaknesses? How does LastPass compare to KeePass? Would you use LastPass?
13. Close all windows.



Project 2-5: Download and Install a Password Generator

Another option to password managers is browser extensions that generate passwords. Instead of storing user-created passwords, these extensions transparently combine multiple elements (such as the username, master password, and site's domain name) into a single site-specific password. The user begins by entering their username and master password, and then the extension generates their site-specific password. The remote site only sees a domain-specific digest instead of the master password itself. In this project you install and use SuperGenPass.

1. Use your web browser to go to **supergenpass.com** (if you are no longer able to access the site through the web address, use a search engine to search for “SuperGenPass”).
2. Drag the icon **SGP** to your browser's bookmarks toolbar.
3. Click **SGP** on your browser's bookmarks toolbar to launch the SuperGenPass dialog box as displayed in Figure 2-6.

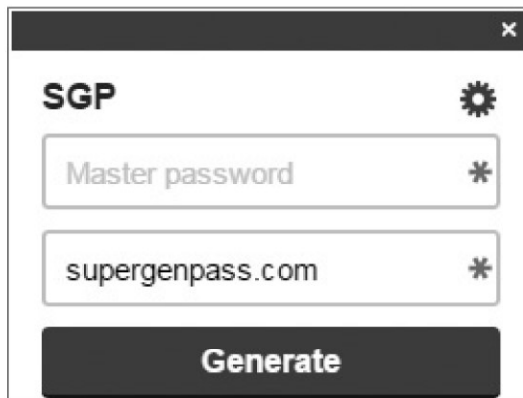


Figure 2-6 SuperGenPass dialog box

Source: SuperGenPass

4. Think of a strong Master Password for SGP but do not enter it yet.
5. Now create an account that requires a password. Go to **gmx.com**.
6. Click **Sign up**.

7. Enter the requested information to create an account. When prompted for a password do not enter a password in GMX but instead click **SGP** in your browser's bookmarks toolbar.
8. Enter your strong **Master password** in the SGP dialog box.
9. Click **Generate**. Note that the password is automatically entered into the password fields.
10. Enter the remaining personal information on the form and create your GMX email account.
11. If necessary, click the **Continue to inbox** button. Click **Log out** to exit your GMX email account.
12. Now access your GMX account with SuperGenPass entering your password. Go to *gmx.com*.
13. Click **Log In**.
14. Enter your email address in the Email box.
15. Click **SGP**.



If nothing happens when you click SGP, you might need to switch to a different browser.

16. The SuperGenPass dialog box opens. Enter your master password created above.
17. Click **Generate**.
18. Click **Log in**.
19. Note that you have entered the website without the need to memorize a unique password for each site.
20. Click **Log out** to exit your GMX email account.
21. How would you rate SuperGenPass compared to KeePass? How does it compare to LastPass? Which of these do you consider the most convenient to use?
22. Close all windows.



Project 2-6: Viewing Your Annual Credit Report

Security experts recommend that consumers receive a copy of their credit report at least once per year and check its accuracy to protect their identity. In this project, you will access your free credit report online.

1. Use your web browser to go to **www.annualcreditreport.com**. Although you could send a request individually to one of the three credit agencies, this website acts as a central source for ordering free credit reports. Figure 2-7 shows the website.
2. Click **Request your free credit reports**.
3. Click **Request your credit reports**. Enter the requested information and click **Next**.
4. Click **TransUnion**. Click **Next**.